

## Four problems, and the scaffolding falls away

$X$  is the input,  $Y$  the middle gene,  $Z$  the output, and every edge is  $+$  (activation) or  $-$  (repression):

$$\frac{dY}{dt} = B_y + f(X, K_{xy}) - \alpha_y Y \qquad \frac{dZ}{dt} = \beta_z G(X, Y) - Z$$

with  $f(u, K) = \frac{(u/K)^H}{1 + (u/K)^H}$  for an activator and  $\frac{1}{1 + (u/K)^H}$  for a repressor. **AND** means

$G = f(X, K_{xz}) f(Y, K_{yz})$ . **OR** means either input alone suffices; the form is

$G = f(X, K_{xz}) + f(Y, K_{yz}) - f(X, K_{xz})f(Y, K_{yz})$ , and you will not need it until the last line of item 2.

Every number in this handout uses  $H = 2$ ,  $K_{xy} = K_{xz} = 0.1$ ,  $\alpha_y = \alpha_z = 1$  and  $\beta_y = 1$ , so time is in protein lifetimes and the input step is  $X : 0 \rightarrow 1$ .  $B_y$  and  $\beta_z$  are stated per item — they are 0 and 1 unless an item says otherwise.

**The sign rule.** Compare the sign of the direct path  $X \rightarrow Z$  with the *product* of the signs along  $X \rightarrow Y \rightarrow Z$ . Same  $\rightarrow$  **coherent**; opposite  $\rightarrow$  **incoherent**.

**The delay convention — state it before you use it.** Response time is the time for  $Z$  to reach **50% of its steady state**; a delay is measured only against simple regulation of  $Z$  by  $X$  alone, with  $\beta_z$  chosen so both reach **the same steady-state**  $Z$ . Then  $t_D = t_{1/2}(\text{FFL}) - t_{1/2}(\text{simple})$ , reported separately for the ON and OFF steps. A delay with no stated comparison and no stated direction is not a number anyone else can check.

### 1 — Fully worked: classify by sign

(a)  $X \xrightarrow{+} Y$ ,  $Y \xrightarrow{+} Z$ ,  $X \xrightarrow{+} Z$ , AND. Direct  $+$ ; indirect  $(+)(+) = + \rightarrow$  **coherent type 1**; the AND gate makes  $Z$  wait for  $Y$ , so **delay on the ON step**.

(b)  $X \xrightarrow{+} Y$ ,  $Y \xrightarrow{-} Z$ ,  $X \xrightarrow{+} Z$ , AND. Direct  $+$ ; indirect  $(+)(-) = - \rightarrow$  **incoherent type 1**:  $X$  drives  $Z$  up and, more slowly, drives up  $Z$ 's repressor. **A pulse, then adaptation**.

(c)  $X \xrightarrow{-} Y$ ,  $Y \xrightarrow{-} Z$ ,  $X \xrightarrow{+} Z$ , AND. Direct  $+$ ; indirect  $(-)(-) = + \rightarrow$  **coherent**, type 4. *Why does that follow?* — a path's sign is the product of its edges.

**Now you:**  $X \xrightarrow{-} Y$ ,  $Y \xrightarrow{+} Z$ ,  $X \xrightarrow{+} Z$ . Coherent or incoherent? \_\_\_\_\_

## 2 — Last step blank: the delay table

Circuit (a), with  $Y_{\min} = 0$ ,  $Y_{\max} = 1$ . On an ON step  $Z$  stays off until  $Y$  crosses  $K_{yz}$ ; since  $Y(t) = Y_{\max}(1 - e^{-\alpha_y t})$ , setting  $Y = K_{yz}$  gives

$$t_D = \alpha_y^{-1} \ln \left[ \frac{Y_{\max} - Y_{\min}}{Y_{\max} - K_{yz}} \right] \quad \text{so at } K_{yz} = 0.5, t_D = \ln 2 \approx 0.69.$$

On an OFF step the AND gate opens the moment  $X$  drops, so  $t_D = 0$ . That formula is the **sharp-gate** ( $H \gg 1$ ) limit. Ours run at  $H = 2$ .

Write your convention here, then fill the two blank columns.

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| $K_{yz}$ | $t_D$ , FORMULA      | $t_D$ , SIMULATED AT $H = 2$ | $t_D$ , OFF STEP |
|----------|----------------------|------------------------------|------------------|
| 0.2      | _____                | 0.35                         | _____            |
| 0.5      | $\ln 2 \approx 0.69$ | 0.75                         | 0                |
| 0.8      | _____                | 1.02                         | _____            |

Why does that step follow? — formula and simulation agree at  $K_{yz} = 0.5$  and disagree badly at 0.8. Which way is the formula wrong there, and what does  $H = 2$  do to the claim that  $Z$  "stays off until  $Y$  crosses"?

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One more line. Same circuit, **OR** gate instead of AND. Which step is delayed now — ON, OFF, or both?

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### 3 — Three blanks: peak, final, and the adaptation error

Circuit (b), and now  $Y$  has a **basal level**:  $B_y = 0.4$ ,  $K_{yz} = 0.5$ ,  $\beta_z = 6$ . Step  $X$  from 0 to 1 and integrate.  $Z$  rises fast, then  $Y$  builds and pulls it back to a plateau that is **not** zero. Both numbers below are *measured off the curve*, not solved for.

peak \_\_\_\_\_ final \_\_\_\_\_

$$\text{adaptation error} = \frac{Z_{\text{final}}}{Z_{\text{peak}}} = \underline{\hspace{2cm}}$$

Why does that step follow? — an error near 0 and one near 1 are two different instruments. What is each good for?

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## 4 — Bare problem

You need a circuit that fires **only if the input has been on for more than about an hour**, and shuts off the moment the input goes away. Every protein you have has a lifetime near 30 minutes.

Which FFL type, and which gate? Then: which parameter would you go after to put the threshold near an hour — and can it actually get you there?

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